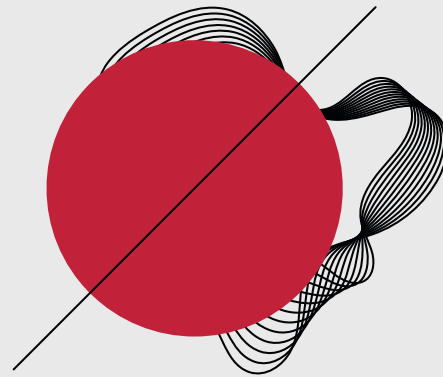


COVID-19 VACCINE RESPONSE

CZ4034 Information Retrieval Group
24



OUR TEAM



Manav Arora



Cheng Xuan



Cheng Lu



**Aksnaya
Muthu**



Parth Taneja





TABLE OF CONTENTS

01. **INTRODUCTION**
Why analyse COVID-19
vaccine response?

02. **CRAWLING**
How we got our data

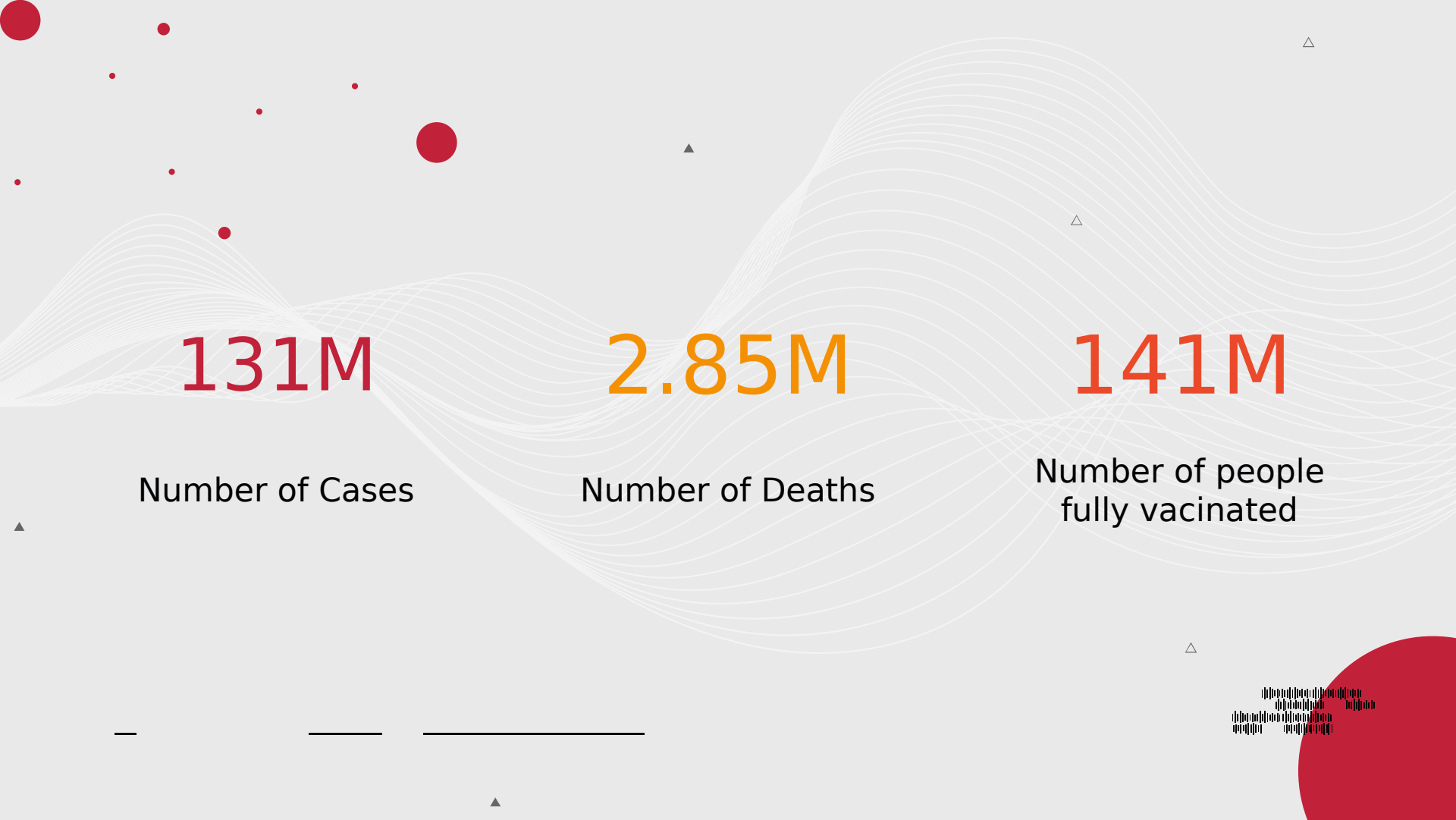
03. **INDEXING &
QUERYING**
UI for Querying

04. **CLASSIFICATION**
Sentiment analysis and
Visualisation of the
data



01.

INTRODUCTION



131M

Number of Cases

2.85M

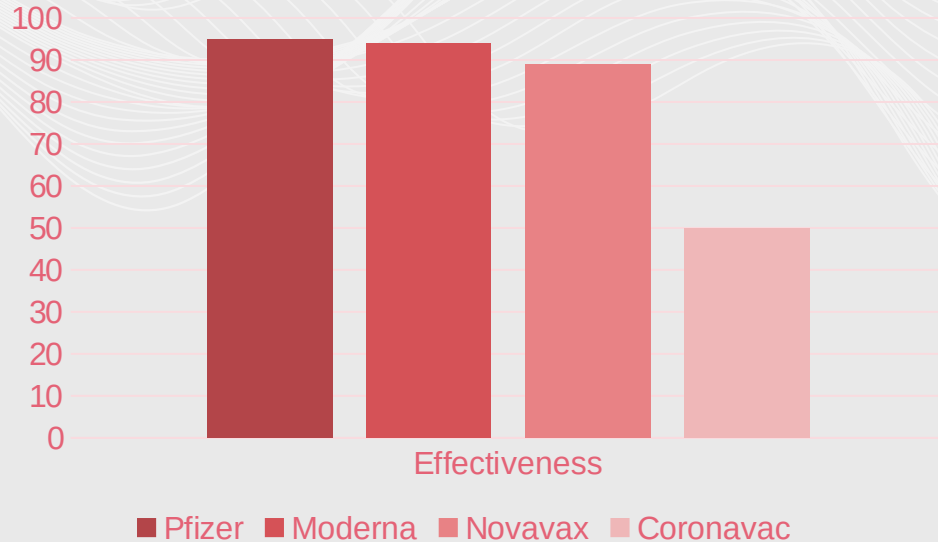
Number of Deaths

141M

Number of people
fully vaccinated



VACCINE EFFECTIVENESS



 **Pfizer**
95%

 **Moderna**
94%

 **Novavax**
89%

 **Coronava
c**
50%



MIXED SENTIMENTS TOWARDS VACCINE .





02.

CRAWLING

60,775
TWEETS
CRAWLED

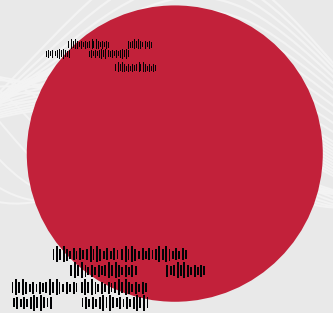
Keywords:

covid vaccine, coronavirus
vaccine, corona vaccine



INFORMATION RETRIEVABLE

- Total number of **Positive, Neutral & Negative** Response
- Tweets containing specific **keyword**
- Tweets from a specific geographic **location**
- Specific **precautionary measures and safety protocols** with respect to the vaccine
- **Approval, effectiveness, access, updates** regarding to vaccine



Records, Words and Types

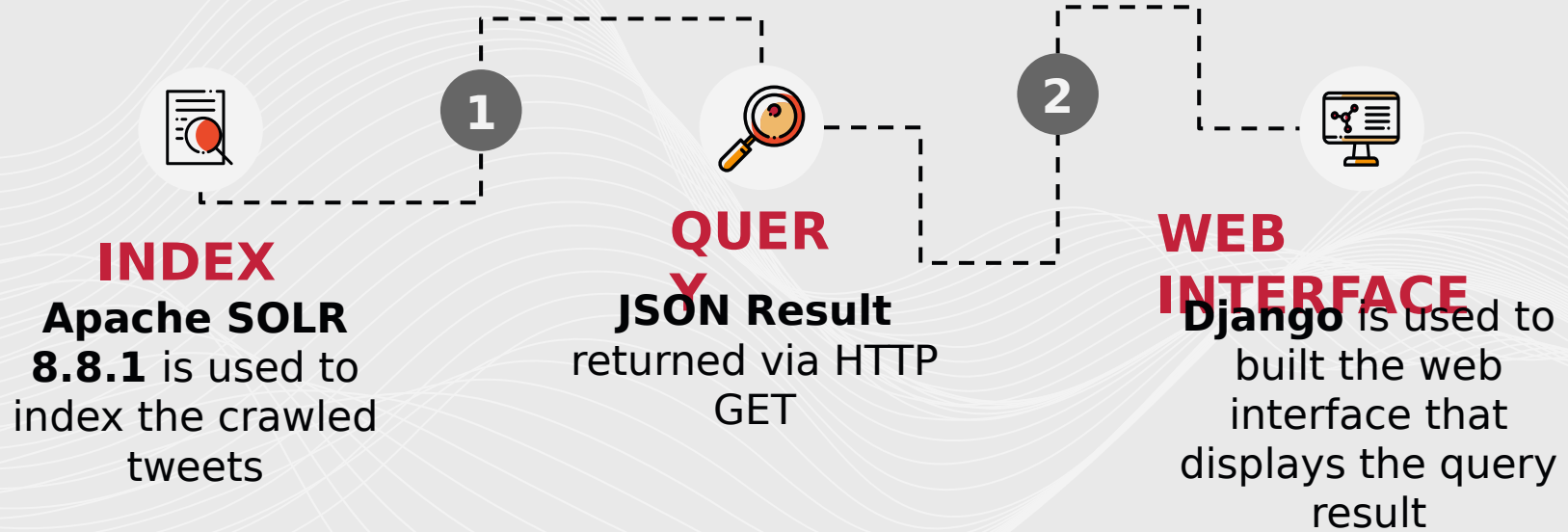
	Total Number of words		Total Number of Unique Words	
	Pre-Processing	Post-Processing	Pre-Processing	Post-Processing
60,775	1,453,183	772,001	159,051	30,237





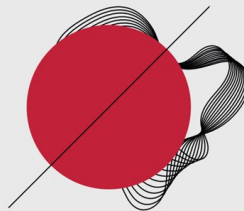
03. INDEXING & QUERYING

Indexing



COVID-19

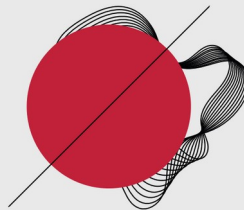
VACCINE RESPONSE



◀ ▶ ✎ 📁 🔍 📄 ☰

COVID-19

VACCINE RESPONSE



— — — — —

Speed of Querying

Query	Speed of Querying (milliseconds)
Processed:hate	0 ms
Processed:"pfizer vaccine"	26 ms
Processed:"worry" OR "hate"	6 ms
Processed:angry^4 covid	1 ms
Processed:happy vaccine~10	1 ms

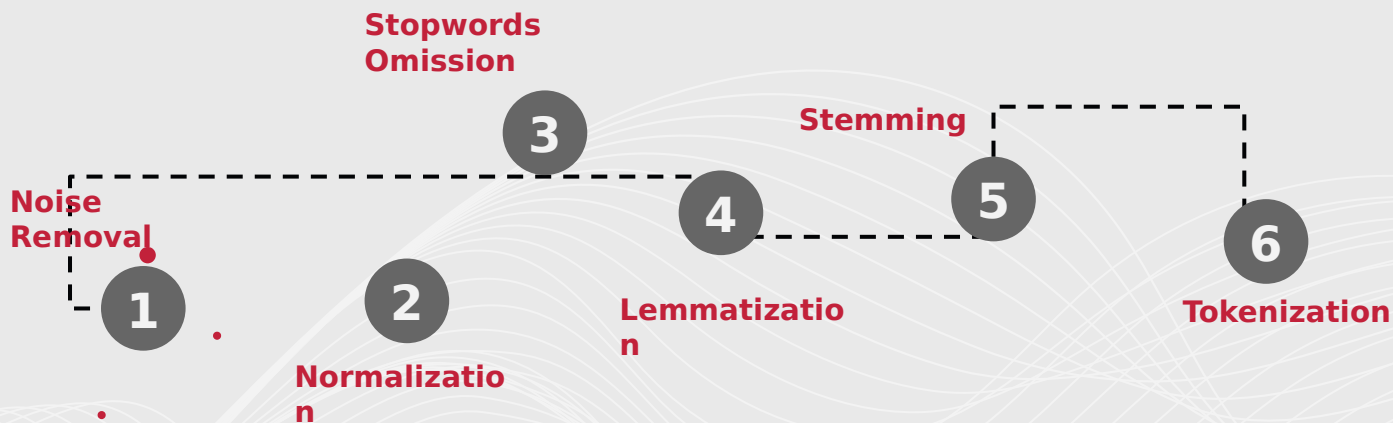




04.

CLASSIFICATION

Pre-Processing



Results of Comparison Models

Model	Count Vectorizer			Tf-idf Vectorizer		
	Precision	Recall	F1 score	Precision	Recall	F1 score
Naïve Bayes	0.824	0.833	0.824	0.812	0.800	0.800
KNN	0.788	0.735	0.735	0.790	0.757	0.757
SVM	0.944	0.944	0.944	0.953	0.953	0.953
Decision Tree	0.801	0.798	0.797	0.790	0.790	0.790

Proposed Models

Bi-LSTM

Model: "sequential"

Layer (type)	Output Shape	Param #
embedding (Embedding)	(None, 182, 50)	1228350
dropout (Dropout)	(None, 182, 50)	0
bidirectional (Bidirectional)	(None, 128)	58880
dropout_1 (Dropout)	(None, 128)	0
dense (Dense)	(None, 3)	387

BERT

Model: "tf_bert_for_sequence_classification"

Layer (type)	Output Shape	Param #
bert (TFBertMainLayer)	multiple	109482240
dropout_37 (Dropout)	multiple	0
classifier (Dense)	multiple	3845
Total params: 109,486,085		
Trainable params: 109,486,085		
Non-trainable params: 0		

Results of Proposed Models

Label	Bi-LSTM			BERT		
	Precision	Recall	F1 score	Precision	Recall	F1 score
Positive	0.86	0.86	0.86	0.96	0.93	0.94
Neutral	0.88	0.94	0.91	0.96	0.98	0.97
Negative	0.90	0.82	0.86	0.94	0.95	0.95
Weighted average	0.876	0.878	0.877	0.952	0.952	0.952

Enhanced Classification

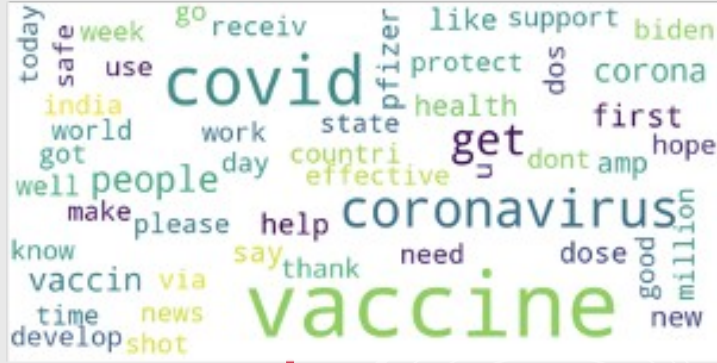
Aspect Based Classification

Model	Metrics		
	Precision	Recall	F1 score
Bi-LSTM	0.876	0.878	0.877
Attention Bi-LSTM	0.903	0.903	0.903

Ensemble Learning

Model	Tf-idf Vectorizer		
	Precision	Recall	F1 score
Decision Tree	0.790	0.790	0.790
Random Forest	0.905	0.904	.903

Visualisation: WordCloud

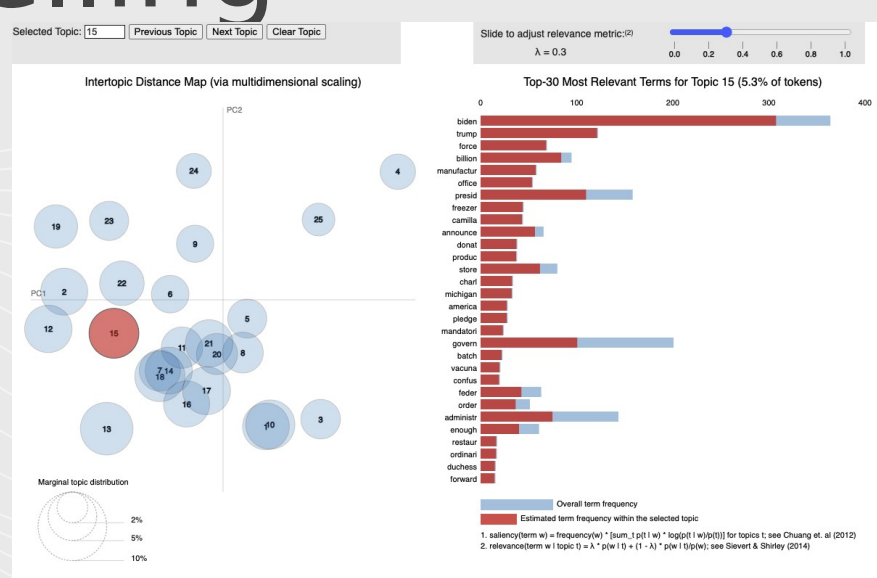
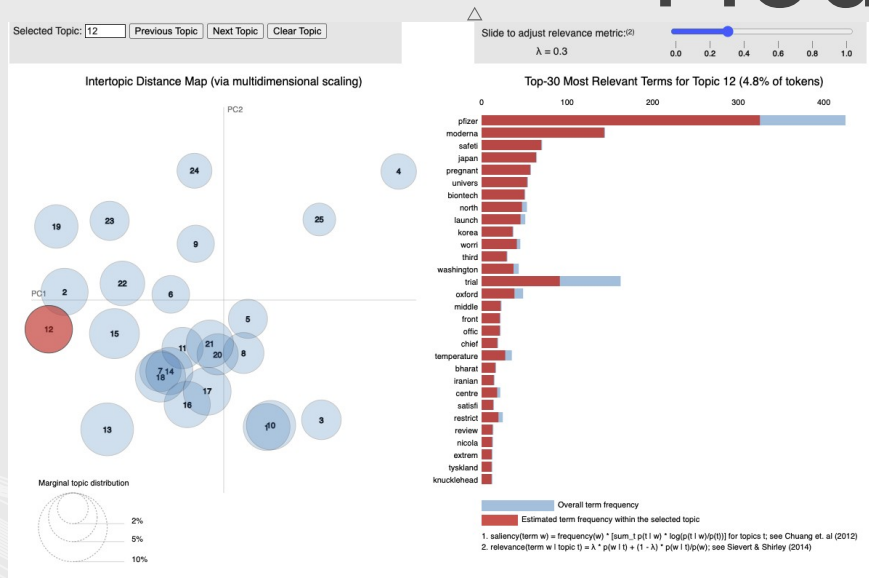


sentiment_type = POSITIVE



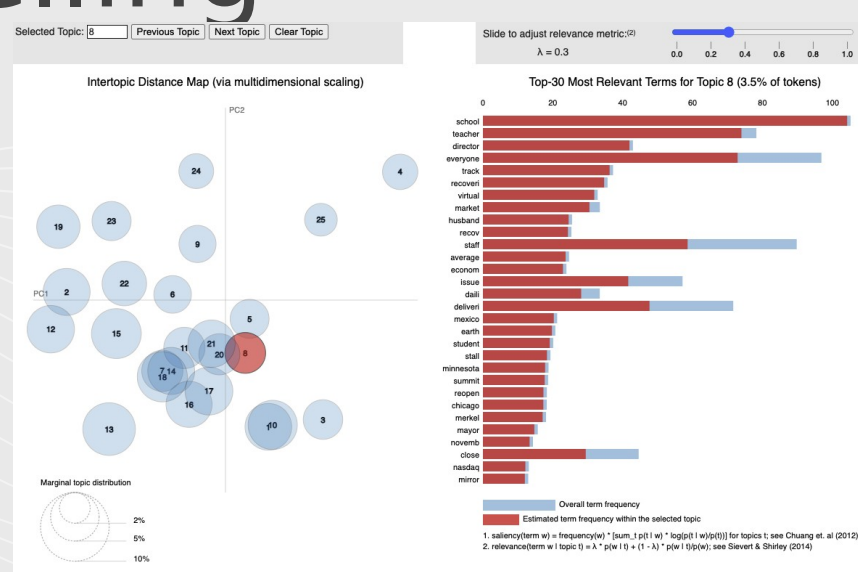
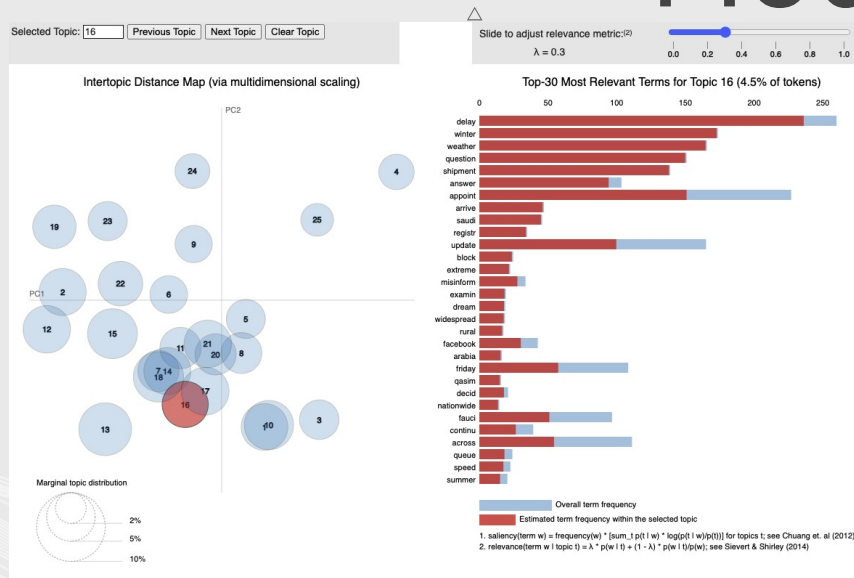
sentiment_type = NEGATIVE

Visualisation: LDA Topic Modelling



“pfizer
r”

Visualisation: LDA Topic Modelling



“delay”

Visualisation: Geospatial Analysis



 POSITIVE



 NEGATIVE

VACCINE TYPES



PROTEIN

Despite being red,
Mars is a cold place



GENE

Jupiter is a gas
giant and the
biggest



HARMLESS

Mercury is the
closest planet to
the Sun

OTHER VACCINES



GAMALEYA

Mercury is the closest planet to the Sun



CANSINO

It's the biggest planet in the Solar System



JANSSEN

Venus is the second planet from the Sun



CORONAVAC

Saturn is the ringed one and a gas giant



NOVAVAX

Despite being red, Mars is a cold place



SINOPHARM

It's the farthest planet from the Sun

A TIMELINE

01.

MERCURY

Mercury is a small planet

02.

JUPITER

Jupiter is the biggest planet

03.

SATURN

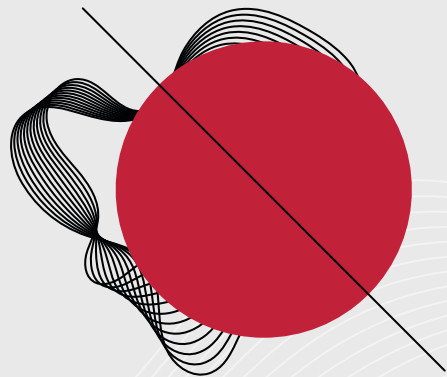
Saturn is the ringed one

04.

NEPTUNE

Neptune is the farthest





AWESOME WORDS



OUR RECOMMENDATIONS

● Write your recommendations here

● Write your recommendations here

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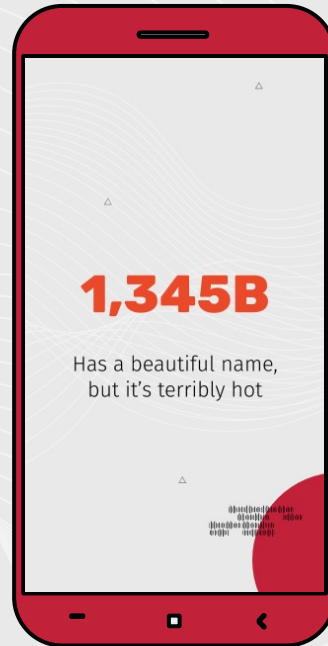
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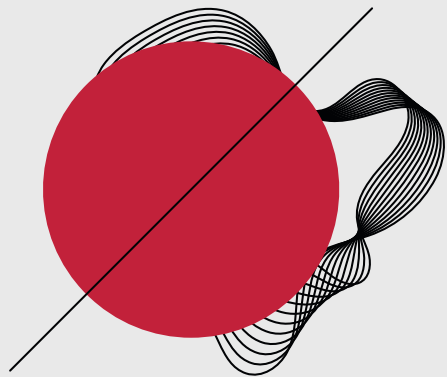
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THANKS!



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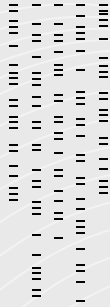
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ICONS





ALTERNATIVE RESOURCES



RESOURCES

VECTOR

- [Geometric background in japanese style](#)
- [Geometric background in japanese style](#)
- [japanese style geometric design](#)

PHOTO

- [Close-up hand holding covid vaccine](#)
- [Doctor vaccinating patient](#)
- [Female scientist with hijab working in lab with microscope](#)
- [Side view of female scientist posing in the lab with microscope and surgical gloves](#)

ICON

- [Research & science Icon Pack](#)



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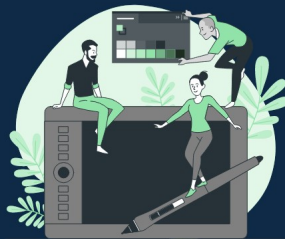
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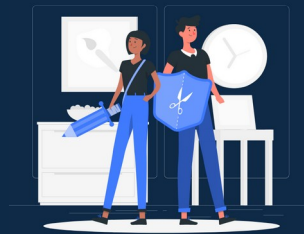
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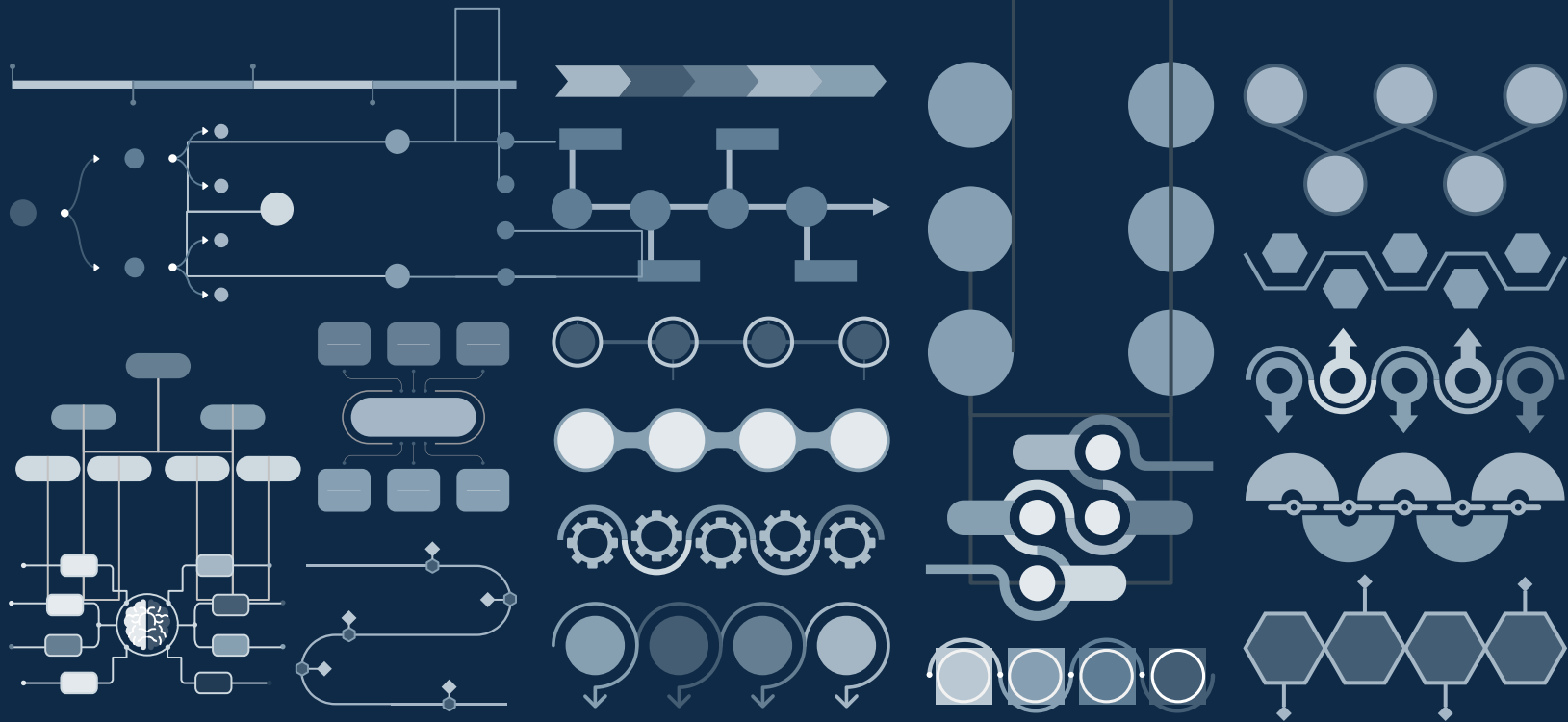
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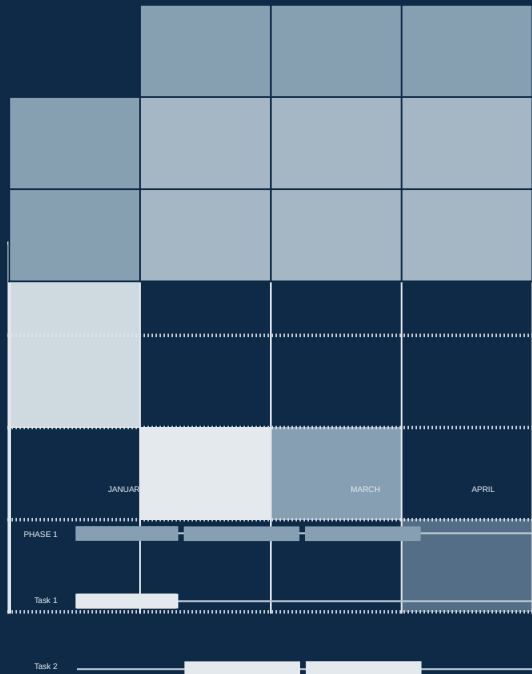
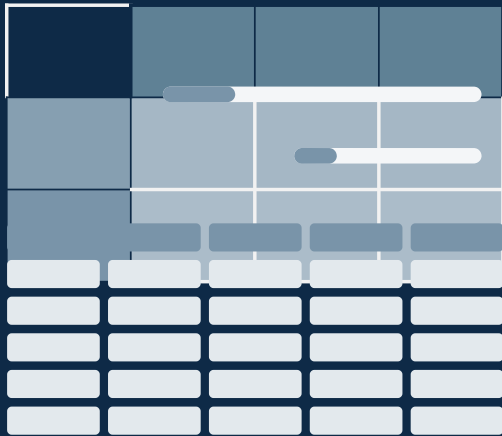
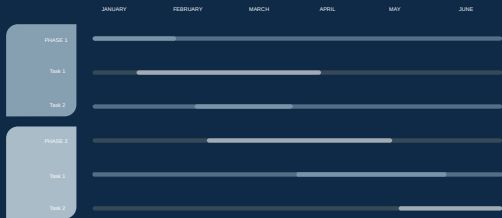
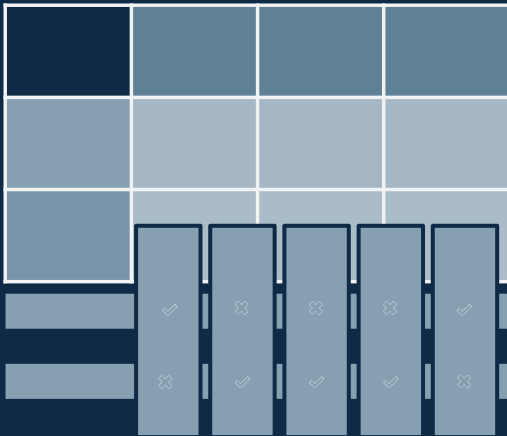
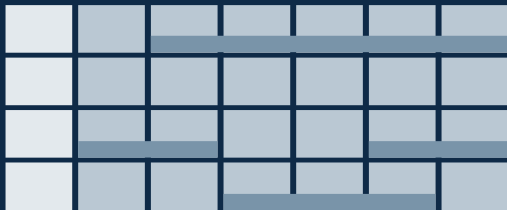
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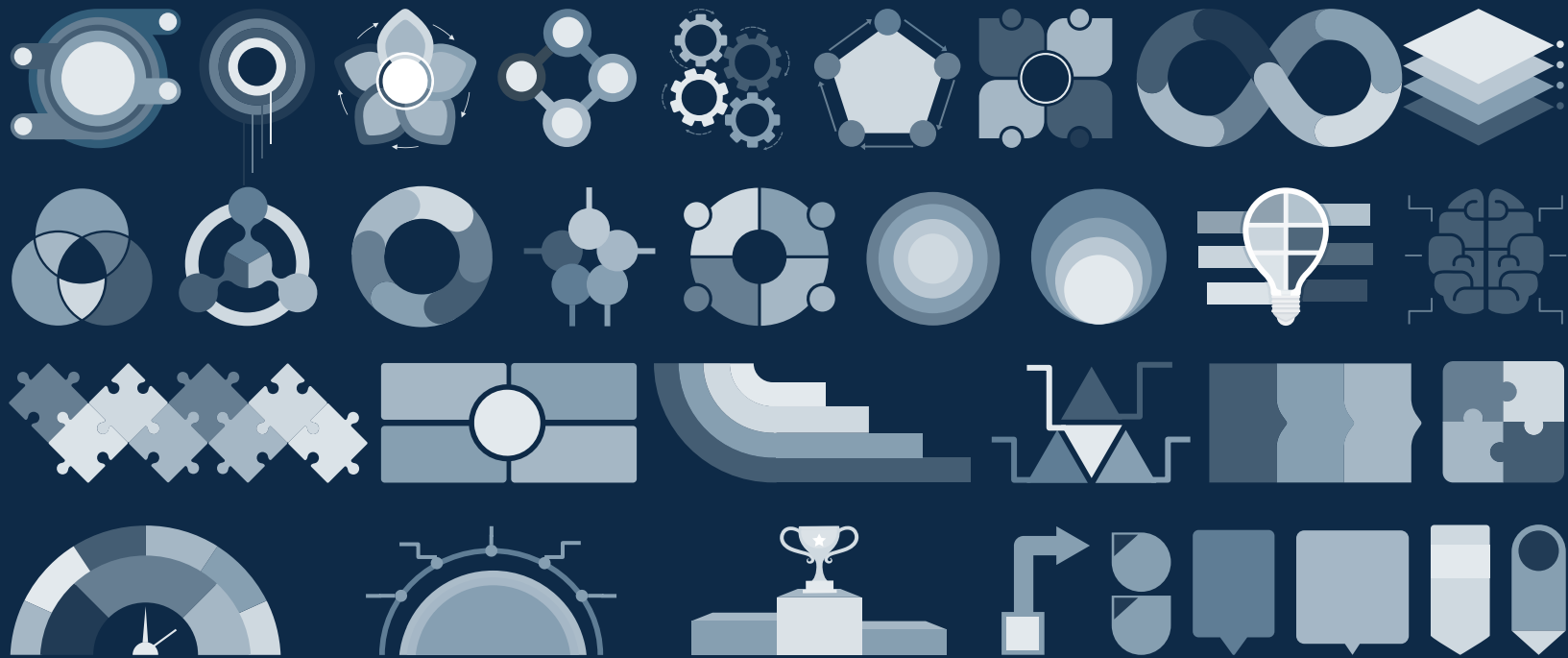
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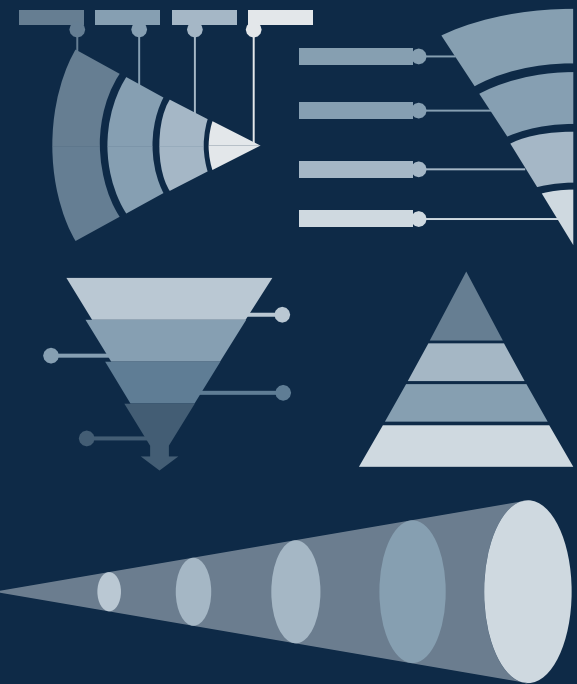
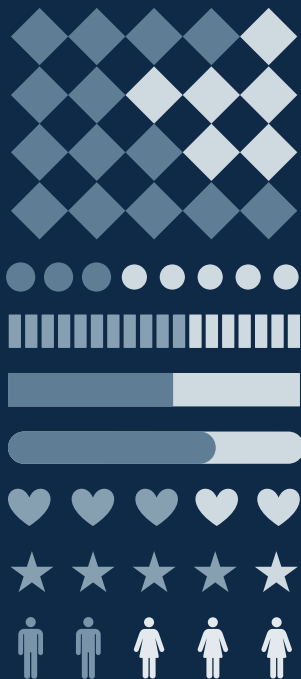
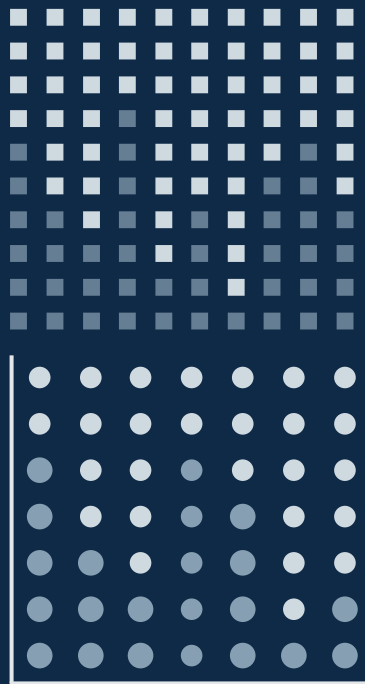




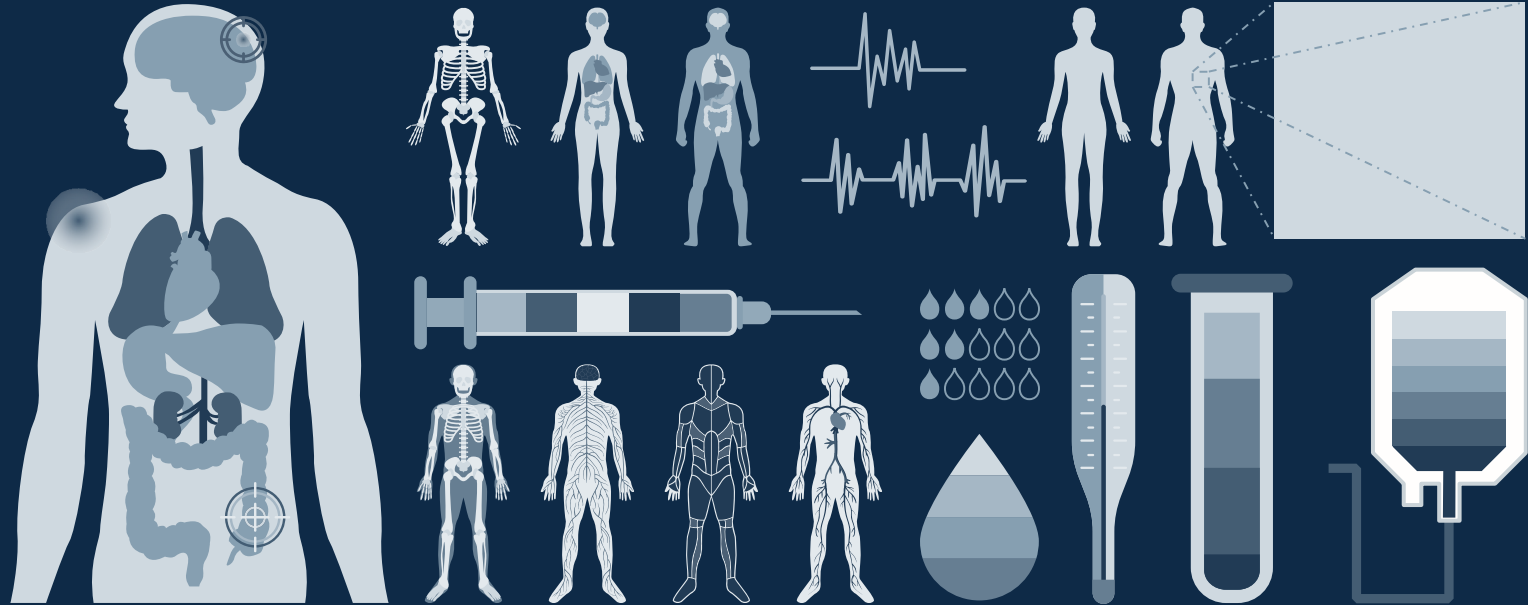


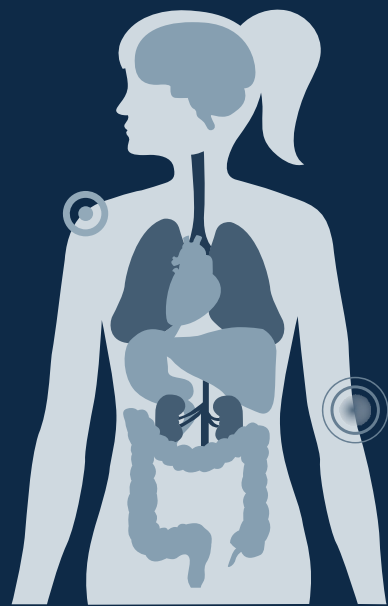


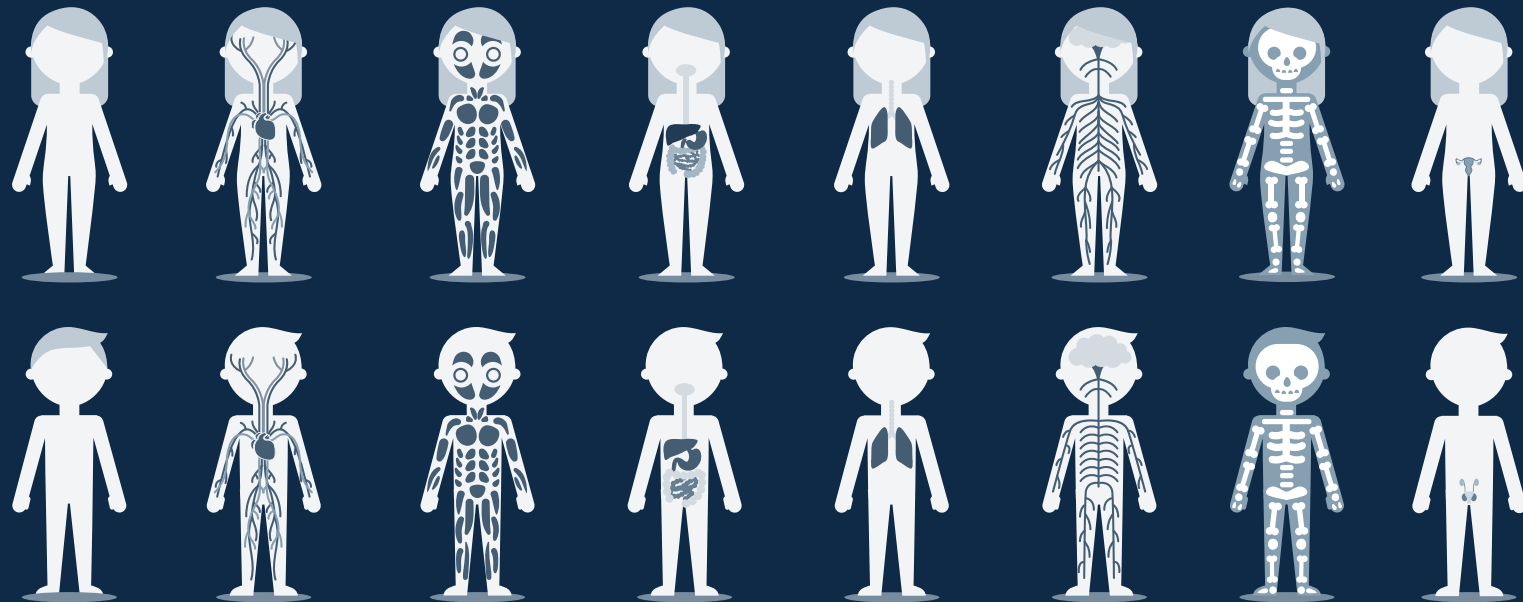




Medical Infographics







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Educational Icons



Medical Icons



A large grid of 100 blue icons representing various business and management concepts. The icons are arranged in a 10x10 grid. The first row includes icons for a person with a speech bubble, a clipboard with a checklist, a gear with a person inside, a clock, two people in a circular flow, two people with a double-headed arrow, a shield with a person, and a globe with a person. The second row features a compass, two hands holding a pen, puzzle pieces, a globe with a circular arrow, a handshake, two people with a speech bubble, a person with a document, and a hierarchical tree diagram. The third row shows a person at a laptop with a speech bubble, a clipboard with a person, two people at a laptop, a document with a person, a cloud with an upload arrow, three people in a hierarchy, three people in a hierarchy, three people in a hierarchy, and three people in a hierarchy. The fourth row depicts a group of people in a circle, a person with a speech bubble, a person at a podium, a video conference screen, a person with a speech bubble, a person on a stage with a flag, a person with a flag, and two hands holding a square. The fifth row contains a lightbulb, a person with a speech bubble, a group of people, a group of people, a globe with people, a folder with a person, a person with a document, and a clipboard with a checklist. The sixth row shows a stopwatch, a magnifying glass over a person, a hand holding coins, a person in a location pin, a chain link, puzzle pieces, a calendar, and a shirt and tie. The seventh row includes a target with people, a bar chart, a bar chart, a presentation board with a graph, a presentation board with a bar chart, a person with a speech bubble, a person with a speech bubble, and a person with a target.

Help & Support Icons



Avatar Icons



Creative Process Icons



Performing Arts Icons



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